

JARVIS

Linux Clustering Monitoring Agent Business Requirements

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1. Executive Summary

The organization requires a Linux Cluster Resource Monitoring System to track hardware utilization across servers in the data center. The primary goal of this system is to provide accurate and timely insights into cluster resource usage to support capacity planning, infrastructure optimization, and operational monitoring.

The system will enable automated collection, storage, and analysis of hardware metrics such as CPU usage, memory utilization, disk I/O, and storage availability across the cluster. These insights will allow the Linux Cluster Administration (LCA) team to proactively manage infrastructure capacity and maintain system performance.

The target implementation date for the monitoring system is January 30, 2026, by which time the system will be evaluated, implemented, tested, and deployed with appropriate documentation and user training.

The solution will include the following capabilities:

- Automated data collection from cluster nodes
- Centralized data storage
- Reporting and analytics capabilities
- Alerting for system failures and high resource usage

Multiple stakeholders will be involved in the project, including:

- Project Manager
- Linux Cluster Administration (LCA) team
- Implementation / Agile development team
- Support and operations team

This document outlines the project objectives, business needs, scope, requirements, stakeholders, timeline, and cost-benefit analysis for the monitoring system implementation.

2. Project Objectives

The primary objectives of the project are:

- Deploy a monitoring agent on all 500 Linux servers within the cluster.
- Collect hardware resource metrics including CPU usage, memory utilization, disk usage, and network statistics.
- Store collected metrics in a centralized database for analysis.
- Generate daily cluster usage reports for the LCA team.
- Provide real-time alerts when servers exceed predefined resource thresholds or become unavailable.

The system must be fully operational and delivering reports by January 30, 2026.

3. Business Needs

The Linux Cluster Administration (LCA) team currently lacks an automated system for monitoring hardware utilization across the cluster. As a result, infrastructure planning and capacity management are inefficient and rely on manual processes or incomplete data.

A monitoring system is required to enable the LCA team to:

- Analyze cluster resource utilization trends
- Identify overloaded or underutilized servers
- Support infrastructure scaling decisions

Based on these insights, the LCA team can add or remove servers to optimize cluster performance and cost efficiency.

Additionally, the monitoring system must support real-time alerting capabilities. The LCA team should receive automated notifications when:

- A server becomes unavailable

- Resource utilization exceeds defined thresholds (e.g., CPU or memory usage above 90%)

This will allow the team to respond quickly to potential performance issues and maintain cluster reliability.

4. Project Scope

Scope management ensures that the project clearly defines the work required and prevents uncontrolled expansion of project requirements (scope creep).

In Scope

The following activities are included in this project:

- Evaluating and selecting technologies for collecting, storing, and analyzing hardware resource data
- Developing and implementing a cluster monitoring system
- Deploying monitoring agents on all Linux servers in the cluster
- Implementing a reporting and analytics system
- Developing a web interface for subscribing and unsubscribing from reports and alerts
- Testing the system in multiple environments including:
 - Development (DEV)
 - Quality Control / Testing (QC)
 - Pre-production (PRE-PROD)
 - Production (PROD)
- Providing training sessions to the support team for ongoing maintenance
- Delivering technical documentation and user manuals to the LCA team

Out of Scope

The following items are not included in this project:

- Monitoring non-Linux infrastructure systems
- Monitoring application-level metrics (only hardware metrics are included)
- Procurement or installation of new physical servers

- Major infrastructure redesign within the data center

5. System Requirements

Based on discussions with stakeholders, the system must meet the following high-level requirements.

Data Collection

- A monitoring agent program must be installed on each server node.
- The agent will collect hardware metrics including CPU, memory, disk usage, and system uptime.
- The agent may be implemented using technologies such as Bash scripts, Python, or Java.

Data Storage

- A centralized database will store collected metrics for analysis.
- Possible database technologies include Relational Databases (RDBMS) or NoSQL databases.

Reporting and Analytics

- The system must support data analysis and reporting through tools such as:
 - SQL
 - Excel
 - Tableau
 - Python-based analytics tools
- Daily cluster usage reports must be generated and delivered before 6:10 PM each day.

Alerting

- Automated alerts must be triggered when:
 - A server becomes unreachable or offline
 - CPU or memory usage exceeds 90% utilization
- Alerts must be delivered through email notifications to the LCA team.

Web Interface

A web-based interface will allow users to:

- Subscribe or unsubscribe from monitoring reports
- Configure alert preferences
- Access monitoring dashboards and reports

6. Key Stakeholders

The following stakeholders are involved in the project implementation.

Project Manager

Responsible for overseeing the project timeline, coordinating stakeholders, and ensuring project deliverables are completed on schedule.

Linux Cluster Administration (LCA) Team

Responsibilities include:

- Providing business requirements related to hardware metrics and reporting needs
- Defining reporting service level agreements (SLAs)
- Consuming cluster monitoring reports and alerts
- Managing infrastructure planning decisions based on monitoring insights
- Funding implementation and ongoing maintenance costs

Implementation / Agile Development Team

Responsible for designing, developing, and testing the monitoring system.

The Agile team consists of:

- Scrum Master
- Product Owner (may also serve as the Project Manager)
- Business Systems Analyst (BSA)
- Two Developers

- One Quality Assurance (QA) Engineer

Support Team

Responsible for:

- Deploying the monitoring system in the production environment
- Installing monitoring agents on all servers
- Maintaining system health and monitoring performance
- Handling system upgrades and ongoing maintenance
- Providing Level 2 support for operational issues

7. Schedule, Timeline, and Milestones

The project will follow the timeline below.

Milestone	Target Date	Responsible Team
Gather business requirements	January 12, 2026	Project Manager / LCA
Develop Minimum Viable Product (MVP)	January 16, 2026	Dev Team
Complete monitoring system implementation	January 22, 2026	Dev Team
Testing in DEV, QC/TEST, PRE-PROD environments	January 24, 2026	QA Team
Deploy system to Production	January 26, 2026	Support Team
Knowledge transfer and training (MCR)	January 27, 2026	Implementation Team / LCA
Deliver first monitoring reports and alerts	January 30, 2026	Support Team

8. Cost-Benefit Analysis

Costs

- Manual labor required to collect hardware usage data from servers
- Infrastructure costs from over-provisioning servers when utilization data is unavailable
- Potential downtime or delays caused by insufficient hardware capacity
- Development, deployment, and maintenance costs of the monitoring system

Benefits

- Improved visibility into cluster resource utilization
- Data-driven infrastructure planning
- Reduced risk of system overload and downtime
- Optimized hardware resource allocation
- Reduced operational costs through efficient server utilization
- Faster response to system failures through automated alerts

Overall, the monitoring system will improve operational efficiency, infrastructure planning, and system reliability, providing significant long-term value to the organization.